Workplace Incidents Analysis
Use case of EU and US

OT7 - Foundation of Data Engineering - 2025 - 2026
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An automated, containerized data pipeline that ingests public occupational accident datasets (ARIA and OSHA), cleans and standardizes them, then builds a star schema for reproducible safety analytics and reporting.

How many times do accidents happen?
Which countries are the most affected?
Which years stands out?
How do EU countries like France compare to the US?

Details

This project analyzes workplace fatal incidents using three heterogeneous datasets (ARIADB, work accidents, fatalities). Because the sources differ in structure, terminology, and available dimensions, we adopt a unified model where each record represents one fatal event (fatality = accident = catastrophe = 1). No event-type distinctions are made; each row counts as a single occurrence.

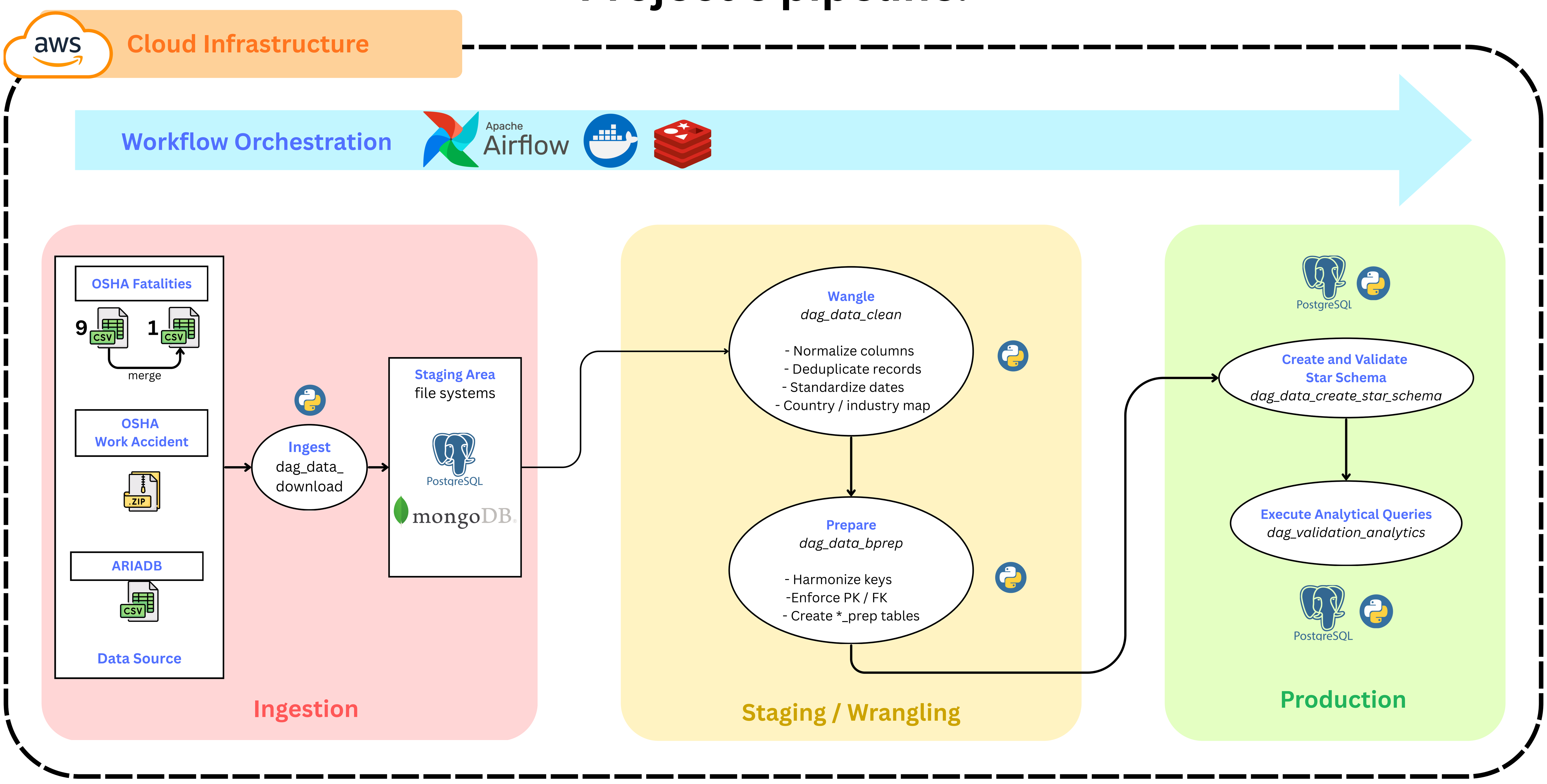
Insight & limitations

Some sources include extra fields like industry or employer, but those dimensions are not consistent across all datasets, so analyses based on them would be partial. The analysis is therefore limited to the common fields: country, date, and fatality count.

Motivation

This approach ensures consistent ETL integration and supports reliable cross-country and time-based comparisons. By favoring completeness and comparability over semantic granularity, the project delivers a coherent analytical view of fatal workplace incidents across all available data.

Project's pipeline:



Datasets used:

OSHA
RÉPUBLIQUE FRANÇAISE
ARIA

GitHub
SCAN ME